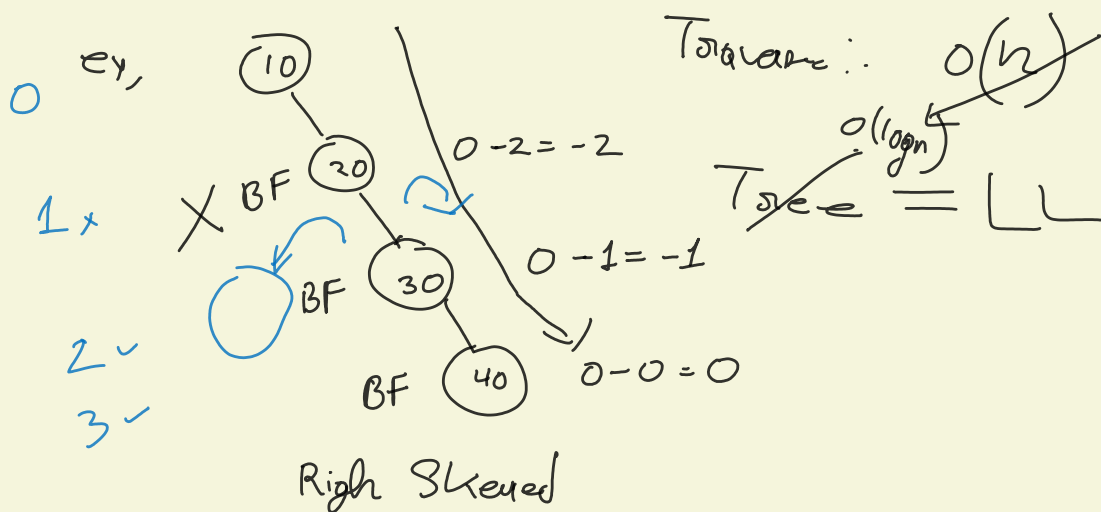
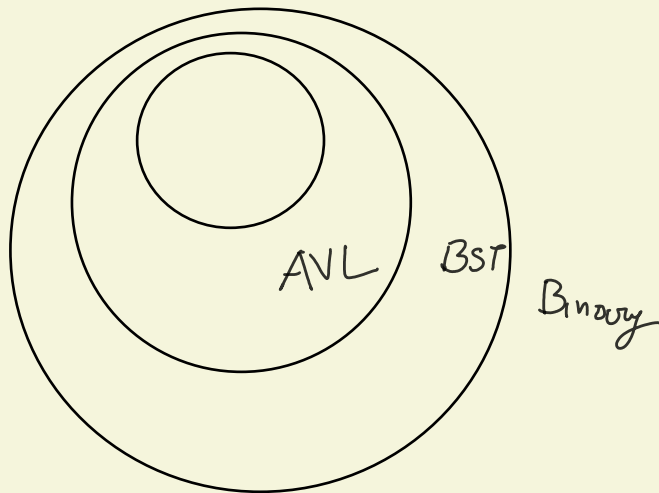


AVL Tree

AVL Tree = Balanced BST



Balancing Factor: $R + = 1$
 $L - = 1$

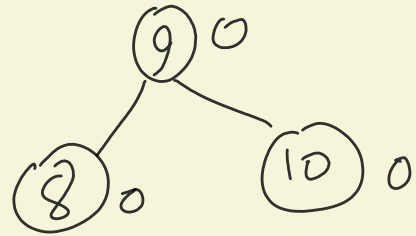
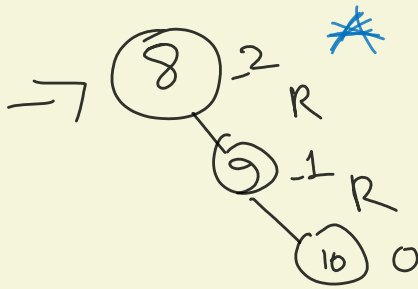
Valid Range of BF = $[-1, 0, 1]$

(Height of Left Sub Tree - Height of Right Sub Tree)

Note: ① BF to be counted from bottom to top

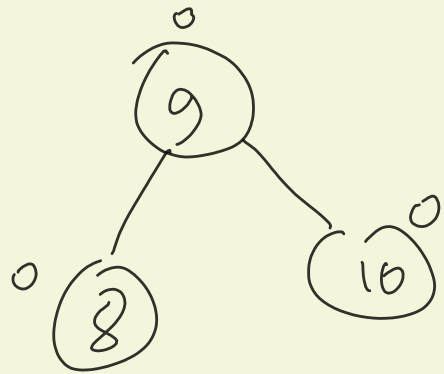
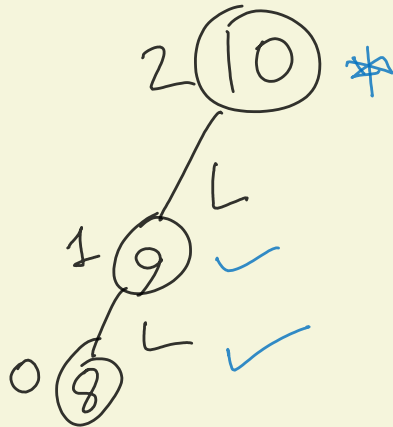
② Rotation has to be done where the BF is out of range.

ex 1
8, 9, 10



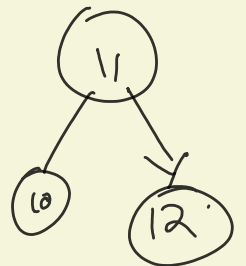
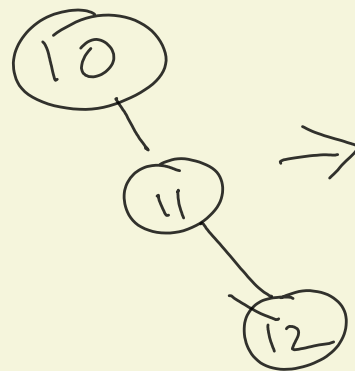
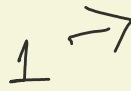
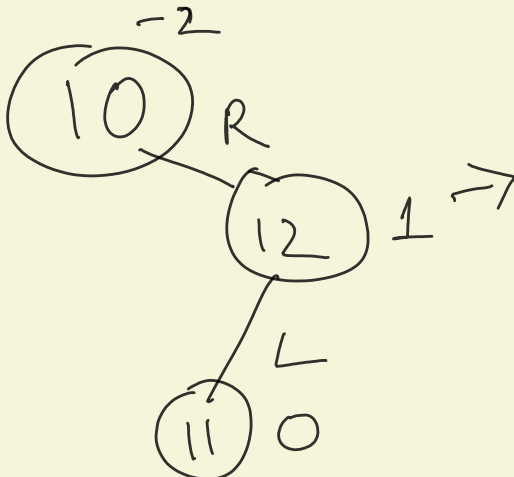
RR → Anti Clockwise rotation or rotate left

ex 2: 10, 9, 8



LL → Clockwise or Right Rotation

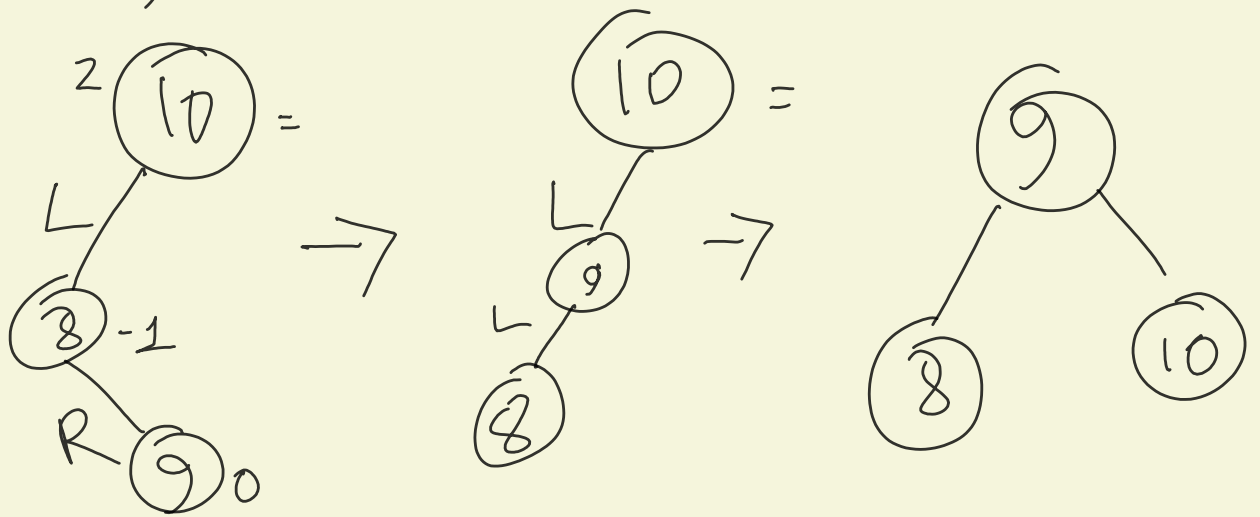
ex 3: 10, 12, 11



L = 0
R = 0

RL → RR → AVL

exq. - 10, 8, 9



LR $\rightarrow$ LL $\rightarrow$ AVL

① How many rotations are needed?

35, 50, 40, 25, 30, 60, 80, 20, 15